Summary:

Erd diagram constraints:

- Mandatory or optional: is every member in superclass participating as a member of a subclass
- Disjoint or nondisjoint: whether member of a superclass can be a member of one, or more than one, subclass (disjoint - or, nondisjoint - and)
    - Disjoint: no overlap
    - Nondisjoint: overlapping (member of more than one subclass)

Converting ERD to RDM

- 1 to many: the 1 is the parent - copy the primary key attributes of the parent into the child relation <u>as a foreign key</u>
- Many to many: create a third relation with two foreign keys to link - primary key is a combination of the primary keys from the original two relations (separated by a comma, both should be underlined)
- 1 to 1: create a third relation, linking ALL ATTRIBUTES - whichever relation starts the action is the primary key, BUT if the first pair of numbers are not 1:1:

  0..1 ; 1..1

  0..1

  Participation / cardinality

  0: optional participation

  Non-zero: mandatory participation

  Mandatory one is parent (1); optional is child (0)
- Recursive relationship: focus on the first pair of numbers

  If both sides are mandatory, only one relation is needed

  If one is mandatory and one is optional, two relations & copy the original primary key twice, change the second copy to be unique

  If both sides start with 0, two relations
- superclass/subclass

  {mandatory, and} - can create a new attribute to identify other tables, instead of creating an entire new table

  {optional, and} - one relation for superclass, one for all subclasses; include attributes from each subclass; include all superclass attributes; create a new attribute to distinguish between the subclasses

  {mandatory, or} - create combined relations for each subclass with their superclass

  {optional, or} - one relation for superclass and each subclass

Normalization: <u>remove the one on the right side</u>

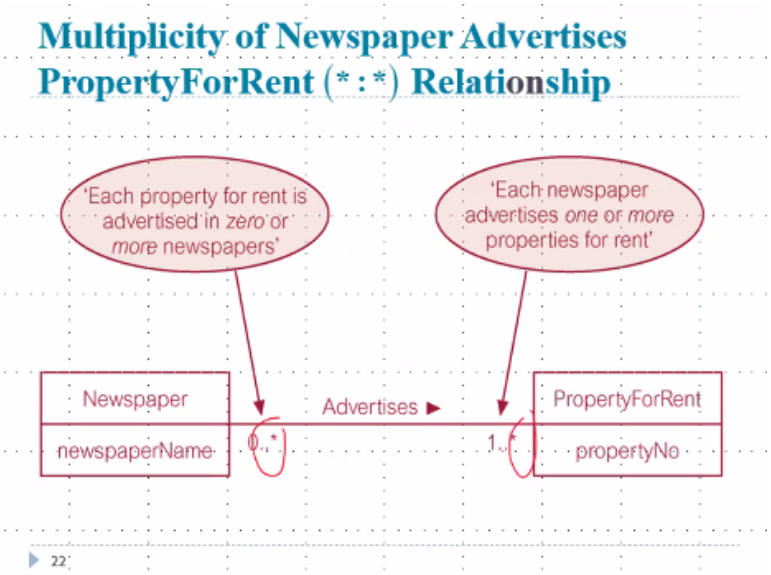
Step 1 key

Step 2 can part of the key determine non key attributes by itself?

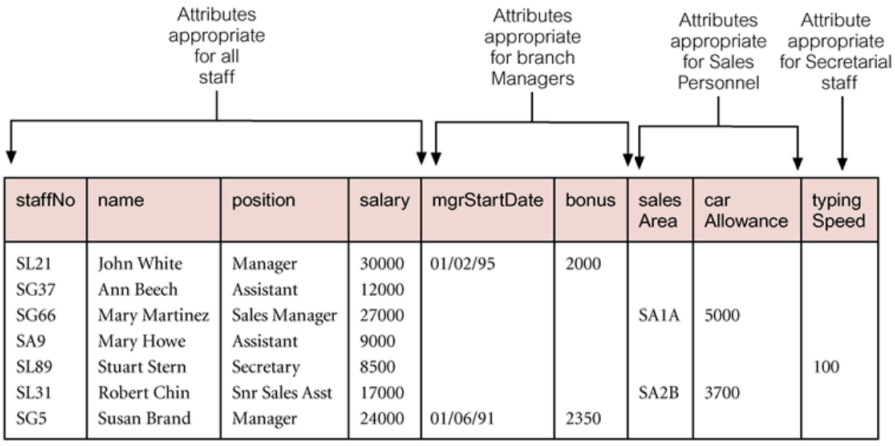
Step 3 can a non key determine a non key?

Step 4 can a non key determine part of a key?

More In depth explanation & visuals:



AllStaff Relation Holding Details of All Staff



staffNo	name	position	salary	mgrStartDate	bonus	sales Area	car Allowance	typing Speed
SL21	John White	Manager	30000	01/02/95	2000			
SG37	Ann Beech	Assistant	12000					
SG66	Mary Martinez	Sales Manager	27000			SA1A	5000	
SA9	Mary Howe	Assistant	9000					
SL89	Stuart Stern	Secretary	8500					100
SL31	Robert Chin	Snr Sales Asst	17000			SA2B	3700	
SG5	Susan Brand	Manager	24000	01/06/91	2350			

Constraints on Specialization/Generalization (2 of 3)

- Participation constraint
  - Determines whether every member in superclass must participate as a member of a subclass.
  - May be **mandatory or optional**.
- Disjoint constraint
  - Describes relationship between members of the subclasses and indicates whether member of a superclass can be a member of one, or more than one, subclass.
  - May be **disjoint or nondisjoint**.
    - "Or" VS "And"

(optional, or)

Disjoint: no overlap
Nondisjoint: overlapping (member of more than one subclass)

Converting ERD Diagram to RDM

Step 2.1 Derive Relations for Logical Data Model

- **One-to-many (1:*) binary relationship types**
  - For each 1: * binary relationship
    - the entity on the 'one side' of the relationship is designated as the parent entity and the entity on the 'many side' is designated as the child entity.
    - To represent this relationship, post a copy of the primary key attribute(s) of parent entity into the relation representing the child entity, to act as a foreign key.

Many to many: must create a third table with two foreign keys to link
Primary key is a combination of the primary keys from the first two tables (separated by a comma, both should be underlined)

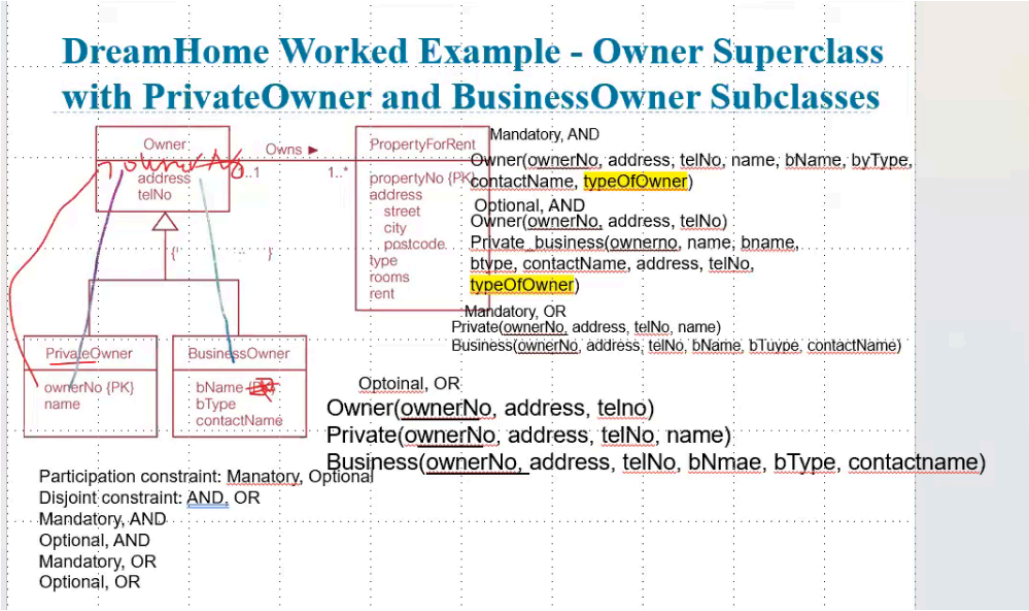
ONE TO ONE
Must also create a third table, linking all attributes
Whichever side starts the action is the primary key for this new relation
BUT, if the first pair of numbers are not 1:1:
0..1 ; 1..1
0..1
Participation / cardinality
0: optional participation
Non-zero: mandatory participation
Mandatory one is parent (1); optional is child (0)

Recursive Relationship
Focus on first pair of numbers
If both sides are mandatory, only one relation is needed
If one is mandatory and one is optional, two relations & copy the original primary key twice, change the second copy to be unique
If both sides start with 0, two relations

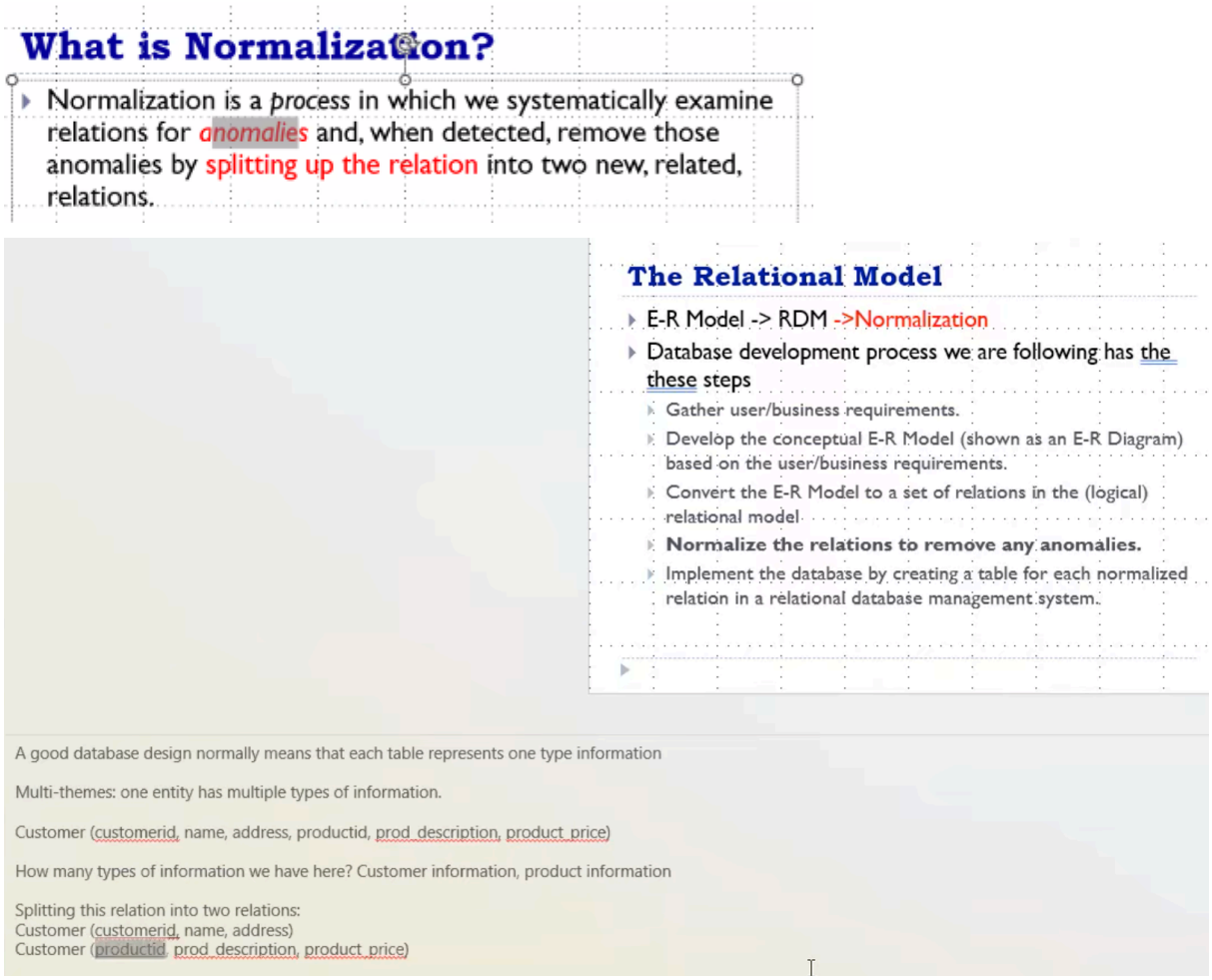
super/subclass

Participation Constraint	Disjoint Constraint	Relations Required
Mandatory	Nondisjoint {And}	Single relation (with one or more discriminators to distinguish the type of each tuple)
Optional	Nondisjoint {And}	Two relations: one relation for superclass and one relation for all subclasses (with one or more discriminators to distinguish the type of each tuple)
Mandatory	Disjoint {Or}	Many relations: one relation for each combined superclass/subclass
Optional	Disjoint {Or}	Many relations: one relation for superclass and one for each subclass

{mandatory, and} - can create a new attribute to identify other tables, instead of creating an entire new table
{optional, and} - one relation for superclass, one for all subclasses; include attributes from each subclass; include all superclass attributes; create a new attribute to distinguish between the subclasses
{mandatory, or} - create combined relations for each subclass with their superclass
{optional, or} - one relation for superclass and each subclass

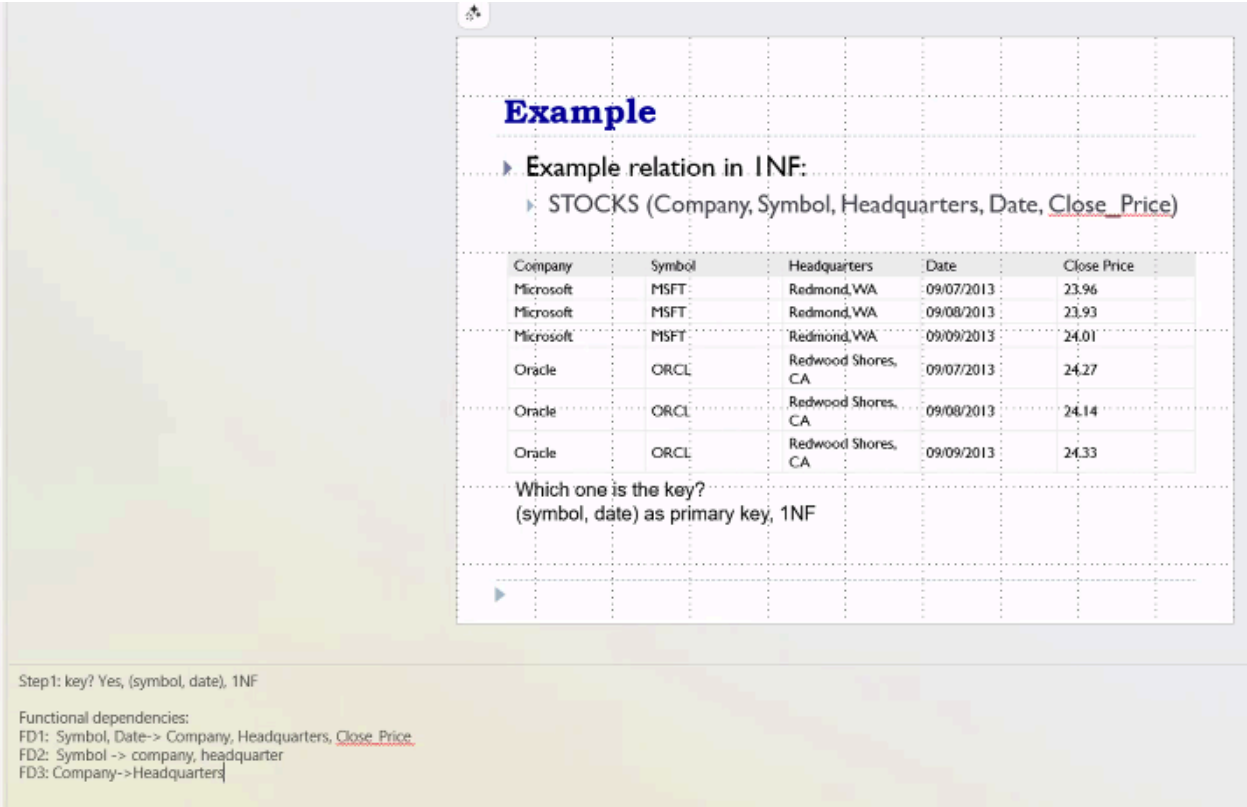


Multi-themes: one entity has multiple types of information
Split the relation into multiple relations to divide the types of info

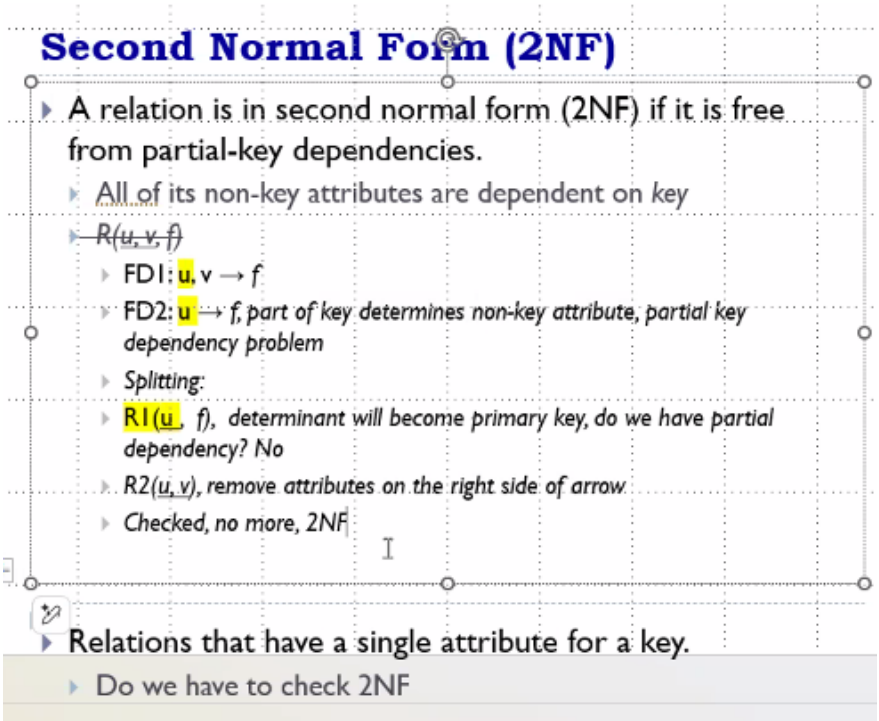


customer(customerid, name, address)
product(product_id, prod_description, prod_price)

First normal form: key



Second normal form: free from partial-key dependencies



Partial key: part of a composite primary key
When part of the key can determine non-key attributes

Third Normal Form (3NF)

- A relation is in third normal form (3NF) if it is in second normal form and it contains no *transitive dependencies*.
- Consider relation R containing attributes A, B and C.

R(A, B, C)

If A → B and B → C then A → C

FD1: A-> B, C

FD2: B->C, non-key attribute determines non-key attribute, we have transitive dependency problem

R1(B, C),

R2(A, B)

checked, no more, 3NF

Transitive dependencies: when non-key attributes can identify non-key attributes
To fix this, you'd move the attribute to a separate table

Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form (BCNF)

- A relation is in BCNF if every determinant is a candidate key.
- Recall that not all determinants are keys.
- Those determinants that are keys we initially call *candidate keys*.
- when non-key attribute -> part of key (primary key must be a composite keY)
- ~~R (a, b, c, d)~~
- FD1: a, b-> c, d
- FD2: c-> a, non-key attribute -> part of key
- R1(c,a)

R 2(b, c, d)

When a non-key attribute can identify part of a key
Always remove the one on the right side

Example

- Example relation in 1NF:
  - ~~STOCKS (Company, Symbol, Headquarters, Date, Close_Price)~~

Company	Symbol	Headquarters	Date	Close Price
Microsoft	MSFT	Redmond, WA	09/07/2013	23.96
Microsoft	MSFT	Redmond, WA	09/08/2013	23.93
Microsoft	MSFT	Redmond, WA	09/09/2013	24.01
Oracle	ORCL	Redwood Shores, CA	09/07/2013	24.27
Oracle	ORCL	Redwood Shores, CA	09/08/2013	24.14
Oracle	ORCL	Redwood Shores, CA	09/09/2013	24.33

Which one is the key?
(symbol, date) as primary key, 1NF

Functional dependencies:
FD1: Symbol, Date-> Company, Headquarters, Close_Price
FD2: Symbol -> company, headquarter, *part of key determines non-key attribute*
FD3: Company->Headquarters

Step2: Do we have partial key dependency? *part of key determines non-key attribute, YES*

~~R1(Symbol, company, headquarter)~~
R2(Symbol, Date, Close_Price), partial key dependency? Symbol-> close_price, date -> close_price

Checked, no more, 2Nf

Step 3 :Do we have transitive dependency? Non-key determines non-key,

YES, Company->Headquarters

R3 (Company, Headquarters)
R4(Symbol, company),

Checked, no more, 3NF

Step 4: Do we have non-key determine part of key? NO, BCNF

Final relations: R2, R3, R4

Ex:

Items are in certain room of certain building

Attribute Name	Sample Value	Sample Value	Sample Value
ITEM_ID	231134-678	342245-225	254668-449
ITEM_LABEL	HP DeskJet 895Cse	HP Toner	DT Scanner
ROOM_NUMBER	325	325	123
BLDG_CODE	NTC	NTC	CSF
BLDG_NAME	Nottooclear	Nottoclear	Canseefar
BLDG_MANAGER	I. B. Rightonit	I. B. Rightonit	May B. Next

Step 1: Key? Yes, Iterm_id, 1NF

Functional dependencies:
FD1: iterm_id -> iterm_label, room_number, bldg._code, bldg._name, bldg._manager
FD2: Bldg_code-> bldg._name, bldg._manager
FD3: Bldg_name-> bldg._manager

Step 2: Do we have partial key dependency? No, 2NF

Step 3: Do we have transitive dependency? Yes, Bldg_code-> bldg._name, bldg._manager

R4(Bldg_code, bldg._name)

R3(Bldg_name, bldg._manager)

R2 (iterm_id , iterm_label, room_number, bldg._code,)

Checked, no more, 3NF

Step 4: do we have non-key determine part of key? No, in BCNF

Final relations: R2, R3, R4